1.8 Solving Absolute Value Equations

Name _____ Period ____ Date _____

Algebra I 1.8 · Student sheet

I can…

- solve absolute value equations.
- solve equations with two absolute values and check for extraneous solutions.
- write absolute value equations for real-life situations.

Vocabulary · fill in the missing word

Absolute value — |a| is the _____ from a to 0 on a number line

Absolute value equation — an equation with a _____ inside absolute value bars

Extraneous solution — a value found while solving that does not satisfy the _____ equation

Part 1 · Distance on the number line

Question	Number(s)	Absolute value equation
Which numbers are 4 units from 0?		
Which numbers are 3 units from 5?		
Which numbers are 2 units from −1?		

Why do most of these have two answers? _____

Part 2 · Notes: fill in the blanks

|ax + b| = c means ax + b = _____ or ax + b = _____.

Isolate the _____ first.

|x| = −3 has _____, because absolute value is never _____.

Always check for _____ solutions.

Part 3 · Examples (work them with the class)

Example 1

Solve each equation.

a. |x| = 6 b. |x + 3| = 10

Example 2

Solve 3|2x − 1| + 4 = 19.

Example 3

Solve $|x - 4| = |2x + 1|$.

Example 4

A bag of chips is labeled 16 ounces. The actual weight can be 0.5 ounce more or less. Write and solve an equation for the least and greatest weights.

Part 4 · Partner practice

Partner 1

Solve $|x - 2| - 3 = 5$.

Partner 2

How many solutions does $|x + 1| = -4$ have?